University of Cape Town
Department of Computer Science

Computer Science CSC3003S
November Examination, 2022

Read through the entire examination paper before you start writing so that you can answer the easy questions first. The relative importance of a question is indicated by the marks allocated to it. Please write legibly.

This paper consists of **Four Sections (there are 7 pages, including this page):**

Section A: *Advanced Software Design 1*

Section B: *Advanced Software Design 2*

Section C: *Theory of Algorithms*

Section D: *Theory of Computation*

Total Marks: 90	Marks per question are shown in brackets.
Time: 3 hours + 10 minutes reading	The use of calculators is permitted.

Please write your answers in FOUR separate books. Use one book for Advanced Software Design 1, one book for Advanced Software Design 2, and one book each for Theory of Algorithms and Theory of Computation. **Label each of the four books as ASD1, ASD2, ToA and ToC**.

Your attention is drawn to the instructions on your answer book on the front and the back. In particular, "**EVERY CANDIDATE MUST enter the book number and the number of each question answered (in the order in which it has been answered)**" on the table provided on the front cover of the first book.

Please **do this for the front cover of the different books used for each section.**

This is needed so that we can allocate marks and not miss questions. **A list of sub-questions is *not* required.**

NOTE: The "Master Theorem" can be found at the end of the paper.

Section A. [22]

Question 1. **[22]**

Refer to the excerpt below and answer the questions relating to the software development methods:

You are developing a system to support pay-as-you-go access to solar power in low-income residential households. The current system is manual; payments are collected in person and the customer's account is updated via a web portal by the employee receiving the payment. You would like to move to a mobile-based automated payment system. You have already secured a tentative agreement with a voucher partner and have access to their systems, in which customers could purchase vouchers in nearby shops and submit the voucher number directly to the web portal, which needs to be integrated with the voucher provider's system. You hope that the voucher system is the right approach, but your job is to get a working payment system in place. You have a limited budget, but since you already have a manual system in place, the new system is urgent but not critical.

a) What project management approach would you use for this project? Give at least three (3) reasons for your choice, clearly comparing your chosen approach to alternatives. [5]

b) You want to give your customers an idea of how they will interact with the payment system, so you can get some early feedback on what will work for them. Describe and classify the prototype you will develop. State any assumptions and **justify** your answer. [3]

c) Test driven development can be described as test-first development plus refactoring. Define test-first development and refactoring. Explain why combining them is necessary in test driven development. [4]

d) You need to test the functionality of the username and password authentication method. What type of test should you use? Motivate your answer. [3]

e) While you initially planned for your customers to submit voucher numbers through a website, after the first iteration you pivot and implement a USSD-based solution. Which principle of the Agile manifesto does this reflect? Explain the principle. [2]

f) You are considering using open source libraries to support your implementation of the system. What are some of the issues (mention at least three) you need to consider and address before you finalise your decision? [3]

g) You realize that the problem you are solving is a common one and are considering making your code open source. Name two tools you will use to support interaction with users and developers and explain how they will be used. [2]

Section B. [23]

Question 2. **[7]**

Refer to the excerpt below and answer the questions that follow.

One example of a management information system is an interactive software development environment that produces a large amount of shared data. These systems are passive in their operations, while control is the responsibility of the system's components.

a) If you were to propose an architectural design for these types of systems, which architecture would it be? [1]

b) Show how this architecture pattern will be implemented with the aid of a diagram. [6]

Question 3. **[6]**

What is Software Quality Defect (SQD)? State any two types of SQD and draw a diagram depicting the "Part-whole hierarchy" as an example of the composite design pattern.

Question 4. **[5]**

a) Consider the scenario below and answer the questions that follow:

You are a programmer with a firm and are required to manage access to a resource that the firm's application must share. Usually, having multiple instances of the same class would be destructive. Your job is to avoid this situation, which is possible when ensuring access to shared resources.

i. What design pattern is most suitable to address the situation in the scenario? [1]
ii. Explain the design pattern identified in (i). [1]
iii. List any two (2) other design patterns in the same category as the pattern you identified above. [2]

b) "Classes do not often start out with LOW COHESION." Under which "SOLID principle" is this type of situation considered? [1]

Question 5. **[5]**

Draw the appropriate diagram that depicts how the implementation of large systems evolves from monolithic applications to micro-services-based systems.

Section C. [23]

Question 6. **[4]**

a) If an algorithm has a running time as *n* (the number of inputs) gets larger that is *exactly* proportional to g(n) = n + log n, then its computational complexity is (choose one answer only): [1]

i. O(log n)
ii. O(n)
iii. Θ(n)
iv. Θ(log n)
v. Ω(log n)

b) In which class of algorithms is Huffman code construction? Choose one answer only. [1]

i. Brute Force
ii. Divide and Conquer
iii. Space-Time Tradeoffs
iv. Dynamic Programming
v. Greedy Technique

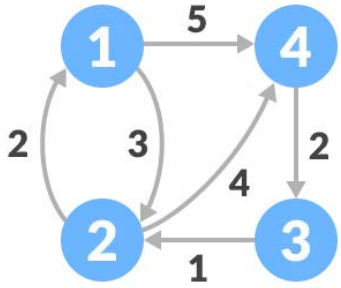
c) For Strassen's multiplication, the number of *additions* for a $n \; x \; n$ matrix is given by the following recurrence relation:

$$A(n) = 7A\left(\frac{n}{2}\right) + 18\,(\tfrac{n}{2})^2 \text{ for } n > 0;\; A(1) = 0$$

Derive the asymptotic growth class for A(n). [2]

Question 7. **[5]**

Given the following directed graph, set up the weight matrix and then apply Floyd's Algorithm for finding the shortest path between every pair of vertices. Show each intermediate matrix.



Question 8. **[4]**

You are given the following string, S, along with a pattern P:

S = SHE_SELLS_SEASHELLS_ON_THE_SEASHORE

P = SHELL

Run Horspool's algorithm for string matching to find the first occurrence of P in S. Show all your working including the shift table and the shifts used.

Question 9. **[3]**

You are given some real value α and asked to compute α^{17}. Use Horner's rule for binary exponentiation to determine *the smallest number of* α *multiply operations* needed for this calculation. Show your working as well as the required table.

Question 10. **[7]**

An array A[0..n − 2] contains n − 1 integers from 1 to n in increasing order, which means that a single integer in the range is missing. For example, 2 is missing from A[] below:

index	0	1	2	3	4	5	6
A[]	1	3	4	5	6	7	8

a) Design an iterative (non-recursive) *decrease by constant* ***factor*** algorithm to find the missing integer with as few comparisons as possible. Be sure to explain the key insights you used in developing this approach and provide a pseudocode implementation. [5]

b) Derive a recurrence relation for the number of key comparisons and from this determine the complexity class of the algorithm. [2]

Section D. [22]

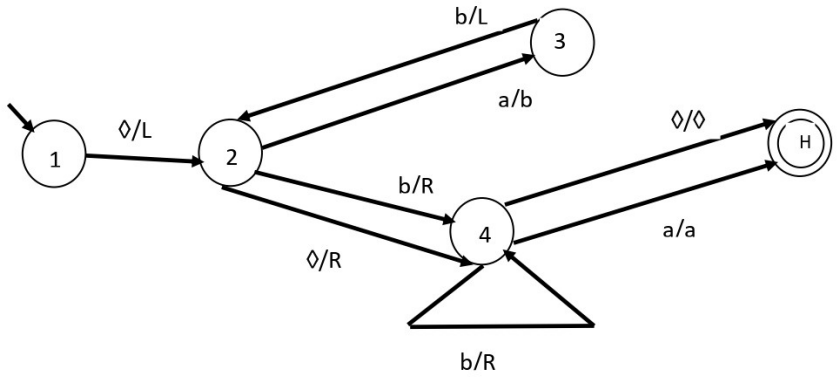
Question 11. **[22]**

a) Explain **ANY TWO** of the following: [2]

i. upper bound
ii. an acceptable language
iii. a nondeterministic Turing machine
iv. the backtracking approach to solving a problem

b) Give a regular expression for the language L over the alphabet of symbols *{A ,B}* where L is the language of all strings that start with an A and end with a B. [1]

c) Consider the Turing Machine below, where ◊ represents a blank cell, L is move Left, and R is move Right:



i. Give any one entry that would appear in the state table for this machine. [2]

ii. Show the output that would result if this machine were given the input below, where the underlined symbol indicates the position of the tape head at the start of execution. Underline the cell in the output to show where the tape head ends up. [2]

◊	a	a	b	a	a	◊	◊	all blank …

iii. What would happen if this machine were given this same input with the tape head positioned at the first cell instead? [1]

◊	a	a	b	a	a	◊	◊	all blank …

iv. What would happen if this machine started executing with the tape head positioned at a cell containing the symbol d ? [1]

d) Use an adversary argument to find a lower bound for the problem of merging two sorted lists. Call their elements A1, A2, …, AN and B1, B2, …, BN. [3]

e) Suppose C and D are problems in class NP (non-deterministic polynomial) and E is NP-complete. Give the numbers of all the statements below which are True:

(Note: $\propto$ indicates *polynomially reducible to*).

1. C is an unsolvable problem
2. D is a tight problem
3. E is in NP
4. if E $\propto$ X and X is in NP, then X is NP-complete
5. if X $\propto$ E and X is in NP, then X is NP-complete [5]

f) Given that the Hamiltonian Path problem ("HP") is known to be NP-Complete, prove that the Travelling Salesman Problem ("TSP") is also NP-Complete. HP asks if there is a tour of an unweighted graph that visits every node exactly once; TSP asks if there is a tour of a weighted graph that visits every node exactly once and has a total weight less than K. [5]

Appendix A – Master Theorem

If we have a recurrence of this form: $T(n) = aT(n/b) + f(n)$, $f(n) \in \Theta(n^d)$ and $T(1) = c$ then

$T(n) \in \Theta(n^d)$ if $a < b^d$

$T(n) \in \Theta(n^d \log n)$ if $a = b^d$

$T(n) \in \Theta(n^{\log_b a})$ if $a > b^d$

-END-